your success = my concern

Tuks.t: 1~2 testing

Pure Mathematics Paper 2: [Time 3hrs]

INSTRUCTIONS TO CANDIDATES

Write your name and surname on the answer paper.

A blue or black pen is required for all the writing.

You may use a pencil for any diagrams or graphs.

Answer all questions in [Section A] and any five questions in [Section B].

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the question, then in the case of an angle it should be given correct to the nearest degree, and in other cases it should be given correct to 2 significant figures.

If a numerical value for g is necessary, take $g = 9.81\text{ms}^{-2}$.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

The total number of the marks for this paper is 120.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

[Turn over]

Section A [40 marks]

Answer **all** questions in this section.

1. Two cirlces C_1 and C_2 have a common chord, PQ, whose equation is $4x + 3y = 36$.

a) Given that the equation of circle C_1 is $x^2 + y^2 - 20x - 14y + 99 = 0$, find the coordinates of P and Q. [6]

PQ is also a diameter of the circle C_2.

b) Find the equation of C_2 in the form $x^2 + y^2 + ax + by + c = 0$, where a, b and c are integers. [4]

c) Find the equation of the tangent to circle C_2 at the point Q. [4]

2a) Using Macluarin's theorem, show by differentiating that a series expansion for $\ln(1 + 2x)$ up to and including the term in x^4 is $\ln(1 + 2x) = 2x - 2x^2 + \frac{8x^3}{3} - 4x^4$. [6]

b) Hence or otherwise find the corresponding Maclaurin's expansion for

$$\ln\left\{\sqrt[3]{\frac{1+2x}{1-2x}}\right\}$$ [3]

3. G is the group of symmetries of an equilateral triangle, under composition of transformations. Its group table is

	a	b	c	d	e	f
a	c	d	a	b	f	e
b	f	e	b	a	c	d
c	a	b	c	d	e	f
d	e	f	d	c	a	b
e	d	c	e	f	b	a
f	b	a	f	e	d	c

a) State the identity element. [1]

b) State whether the element a represents a reflection or a rotation. Justify your answer. [2]

c) Find a subgroup of order 3. [2]

d) Find the period of element b. [1]

e) Explain why the group G has no element of order 4. [1]

f) Is group G cyclic or not? Give a reason for your answer. [2]

4. A circular cylinder is expanding in such a way that, at time t seconds, the length of the cylinder is $20x$ cm and the area of the cross-section is x cm^2. Given that, when $x = 5$, the area of the cross-section is increasing at a rate of $0{,}025\text{cm}^2 s^{-1}$, find the rate of increase at this instant of

a) the length of the cylinder, [2]

b) the volume of the cylinder, [3]

c) the radius of the cylinder. [3]

Section B [80 marks]

Answer any **five** questions from this section. Each question carries **16 marks**.

5a) Let **N** = $\begin{pmatrix} 0 & -1 & 2 \\ 1 & 1 & 2 \\ -1 & p & 1 \end{pmatrix}$.

(i) State the dimensions of matrix **N**. [1]

(ii) Find the rational value of p for which this matrix does not have an inverse. [4]

(iii) If $p = 3$, find the inverse of **N**. [6]

b) The matrix **P** = $\begin{pmatrix} 3 & -1 \\ 0 & 1 \end{pmatrix}$ represents a linear transformation of the $x - y$ plane.

Find the equation of the straight line through the origin, each of whose points is invariant under this transformation. [5]

6a) The plane π_1 has equation $2x - 3y + 6z = 16$. The plane π_2 is parallel to plane $\pi_1$and contains the point with position vector $\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}$.

(i) Find the cartesian equation of π_2. [2]

(ii) Calculate the perpendicular distance between plane π_1 and plane π_2. [3]

b) The straight line l has equation $\mathbf{r} = \mathbf{i} + 6\mathbf{j} - 3\mathbf{k} + s(\mathbf{i} - 2\mathbf{j} + 2\mathbf{k})$. Another plane p has equation $(\mathbf{r} - 3\mathbf{i}).(2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}) = 0$. The line l intersects the plane p at the point A.

(i) Find the position vector of A. [3]

(ii) Find the acute angle between l and p. [3]

(iii) Find a vector equation for the line which lies in p, passes through A and is perpendicular to l. [5]

7a) A certain curve is such that its gradient at a point $(x; y)$ is proportional to xy. At the point (1; 2) the gradient is 4.

(i) By setting up and solving a differential equation, show that the equation of the curve is $y = 2e^{x^2-1}$. [7]

(ii) State the gradient of the curve at the point $(-1; 2)$ and sketch the curve. [2]

b) Liquid is flowing into a small tank which has a leak. Initially the tank is empty and, t minutes later, the volume of liquid in the tank is V cm^3. The liquid is flowing into the tank at a constant rate of 80 cm^3 per minute. Because of the leak, liquid is being lost from the tank at a rate which, at any instant, is equal to kV cm^3 per minute where k is a positive constant.

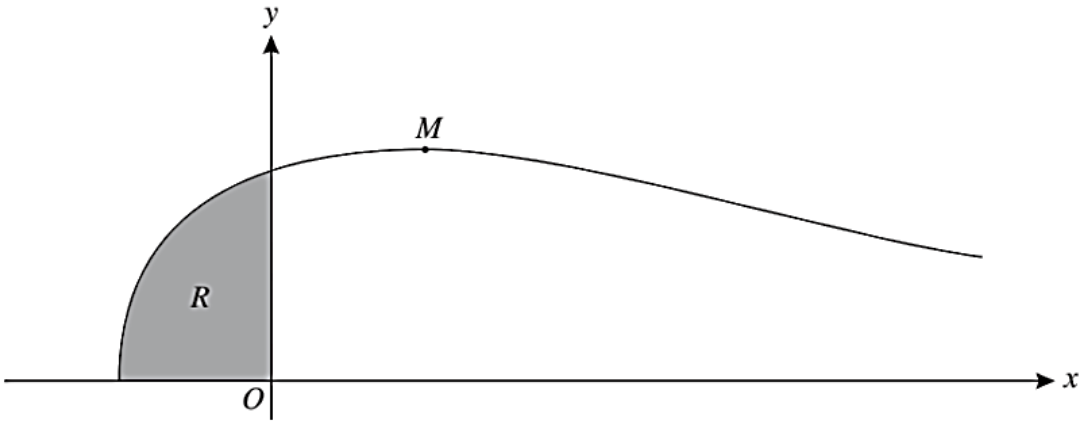
Write down a differential equation describing this situation and solve it, obtaining an expression of V in terms of e, k and t. [7]

8a) (i) Prove the identity $\sin^2\theta\cos^2\theta \equiv \frac{1}{8}(1 - \cos 4\theta)$. [3]

(ii) Hence find the exact value of

$$\int_0^{\frac{1}{3}\pi} \sin^2\theta\cos^2\theta \ \mathrm{d}\theta.$$ [3]

b)



The diagram shows the curve $y = e^{-\frac{1}{2}x}\sqrt{1+2x}$ and its maximum point M. The shaded region between the curve and the axes is denoted by R.

(i) Find the x-coordinate of M. [4]

(ii) Find the volume of the solid obtained when R is rotated completely about the x-axis. Give your answer in terms of π and e. [6]

9a) (i) Find the four roots of the equation $z^4 = 8(\sqrt{3} + \mathrm{i})$ in the form $z = r\mathrm{e}^{i\theta}$. [5]

(ii) Show these roots on an Argand diagram. [2]

b) (i) Use de Moivre's theorem to show that $sin^5\theta \equiv a \sin 5\theta + b \sin 3\theta + c \sin \theta$ where a, b and c are constants to be found. [5]

(ii) Hence, leaving your answer as a fraction, find

$$\int_0^{\frac{\pi}{3}} sin^5\theta \, d\theta.$$ [4]

10a) (i) Show that, for $r > 0$

$$\frac{1}{r^2} - \frac{1}{(1+r)^2} \equiv \frac{2r+1}{r^2(r+1)^2}$$ [1]

(ii) Hence prove that, for $n \in \mathbb{N}$

$$\sum_{r=1}^{n} \frac{2r+1}{r^2(r+1)^2} = \frac{n(n+2)}{(n+1)^2}$$ [3]

(iii) Show that, for $n \in \mathbb{N}, n > 1$

$$\sum_{r=n}^{3n} \frac{6r+3}{r^2(r+1)^2} = \frac{an^2+bn+c}{n^2(3n+1)^2}$$

where a, b and c are constants to be found. [3]

b) (i) Find the Taylor series expansion about $\frac{\pi}{4}$ of $\tan x$ in ascending powers of $\left(x - \frac{\pi}{4}\right)$ up to and including the term in $\left(x - \frac{\pi}{4}\right)^3$. [7]

(ii) Deduce that an approximation for $\tan \frac{5\pi}{12}$ is $1 + \frac{\pi}{3} + \frac{\pi^2}{18} + \frac{\pi^3}{81}$. [2]

11a) Using the principle of mathematical induction, prove that

$$\sum_{r=1}^{n} \frac{3r+2}{r(r+1)(r+2)} = \frac{n(2n+3)}{(n+1)(n+2)} \,. \qquad [6]$$

b) (i) On a single diagram sketch the graph of $y = \pi \sin x$ and $y = \pi - x$, for $-3\pi \leq x \leq 3\pi$, and hence state the number of real roots of the equation $\pi \sin x = \pi - x$. [4]

(ii) Show by calculation, that the equation has a root between $x = 0$ and $x = \frac{\pi}{2}$. [2]

(ii) Taking $\frac{1}{4}\pi$ as a first approximation to one of the roots α, apply Newton-Raphson process once to find a second approximation to α, giving three decimal places in your answer. [4]

End of paper